

Show command palette (Ctrl+Shift+P)

```
import cv2
import matplotlib.pyplot as plt

# Read image
img = cv2.imread('sample.jpg')
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# -----
# Simulate aliasing
# -----
# Downsample the image
small = cv2.resize(img, (100, 100), interpolation=cv2.INTER_LINEAR)

# Upsample using nearest neighbour (creates aliasing)
aliased = cv2.resize(small, (img.shape[1], img.shape[0]),
                    interpolation=cv2.INTER_NEAREST)

# -----
# Correct aliasing
# -----
# Smooth image before resizing
blur = cv2.GaussianBlur(img, (5,5), 0)

# Downsample and upsample using bicubic interpolation
small2 = cv2.resize(blur, (100,100), interpolation=cv2.INTER_LINEAR)
corrected = cv2.resize(small2, (img.shape[1], img.shape[0]),
                    interpolation=cv2.INTER_CUBIC)

# -----
# Display Results
# -----
plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")
```

```
plt.subplot(1,3,2)
plt.imshow(aliased)
plt.title("Aliased Image")
plt.axis("off")
```

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```
plt.subplot(1,3,3),
plt.imshow(corrected)
plt.title("Corrected Image")
plt.axis("off")

plt.show()
```

Original Image



Aliased Image



Corrected Image



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